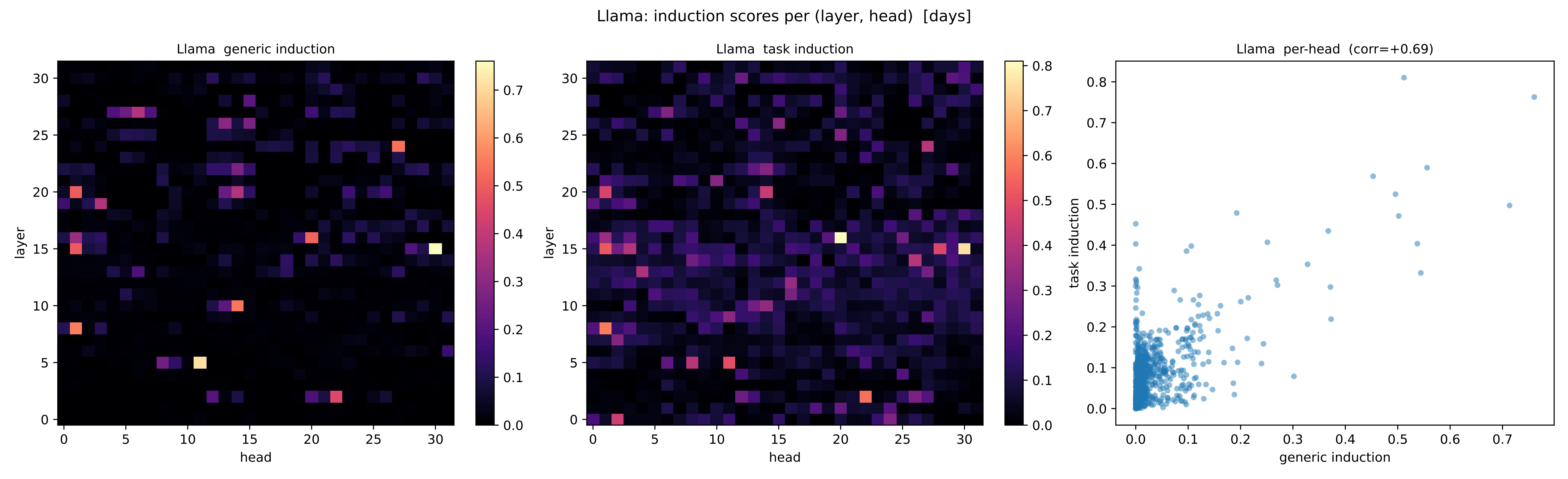
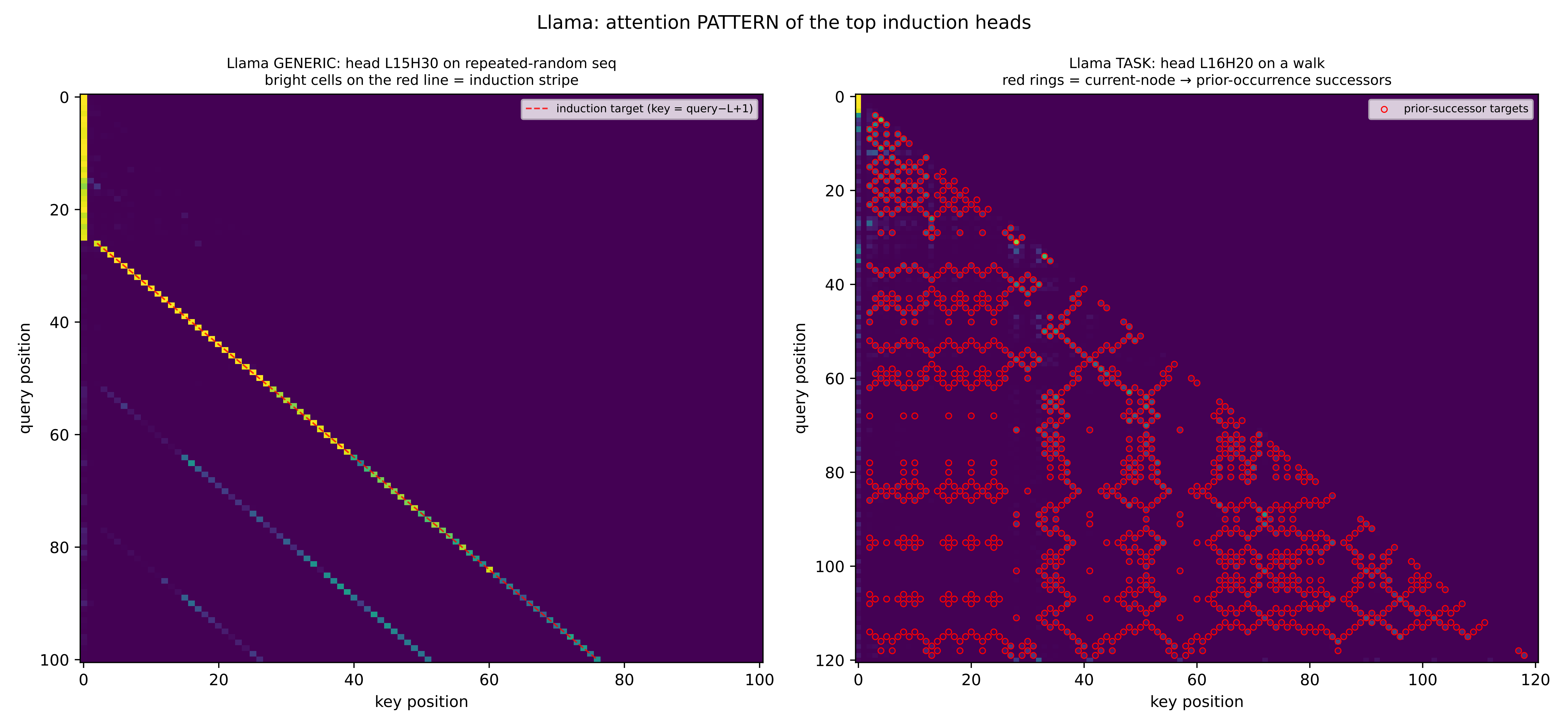
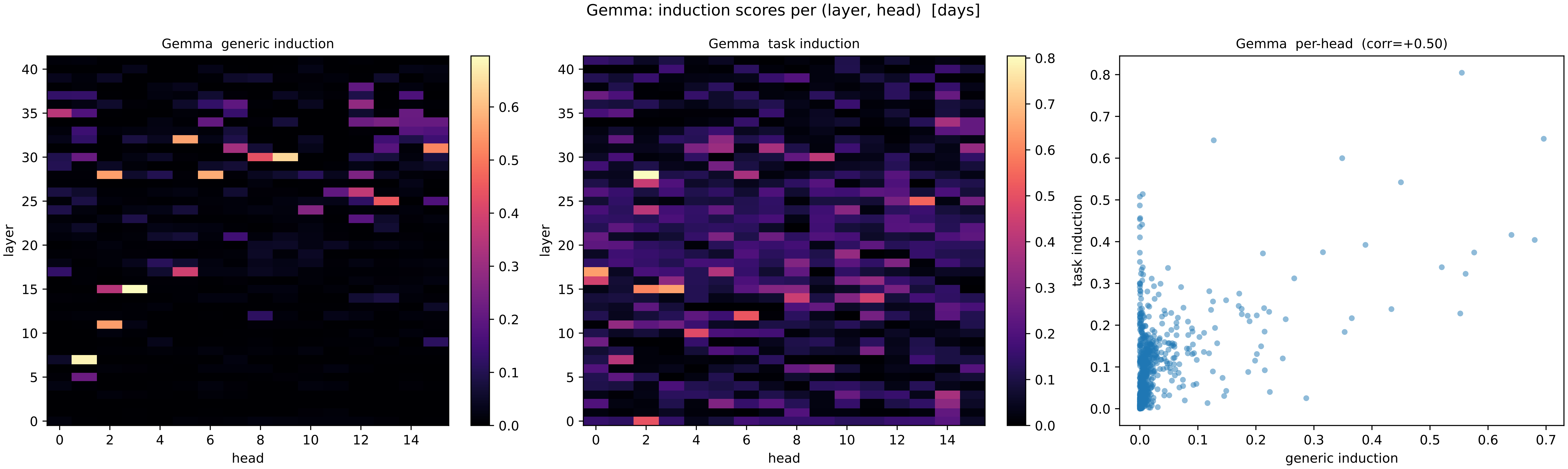


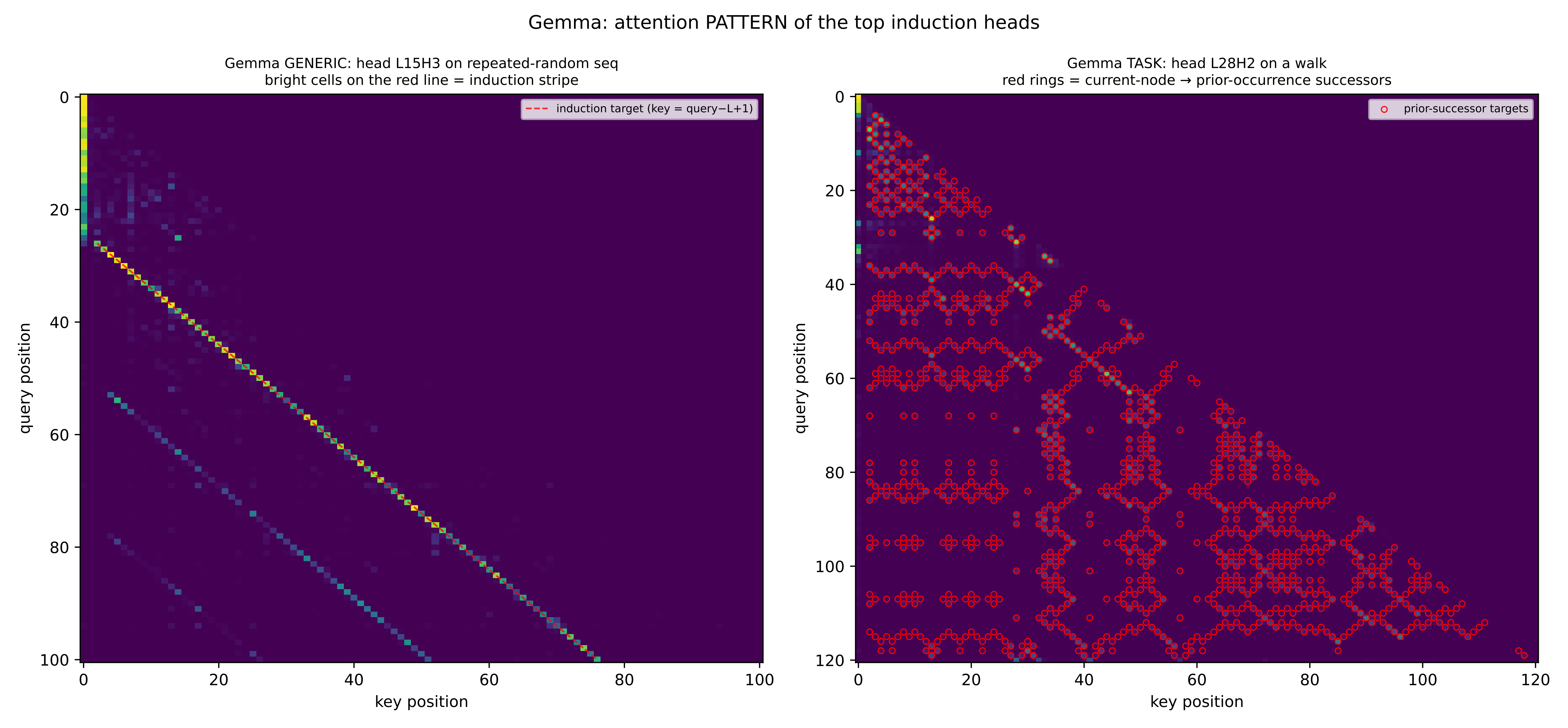
Llama: induction scores per (layer, head) [days]
Llama generic induction
Llama task induction
Llama per-head (corr=+0.69)
layer
head
task induction
generic induction

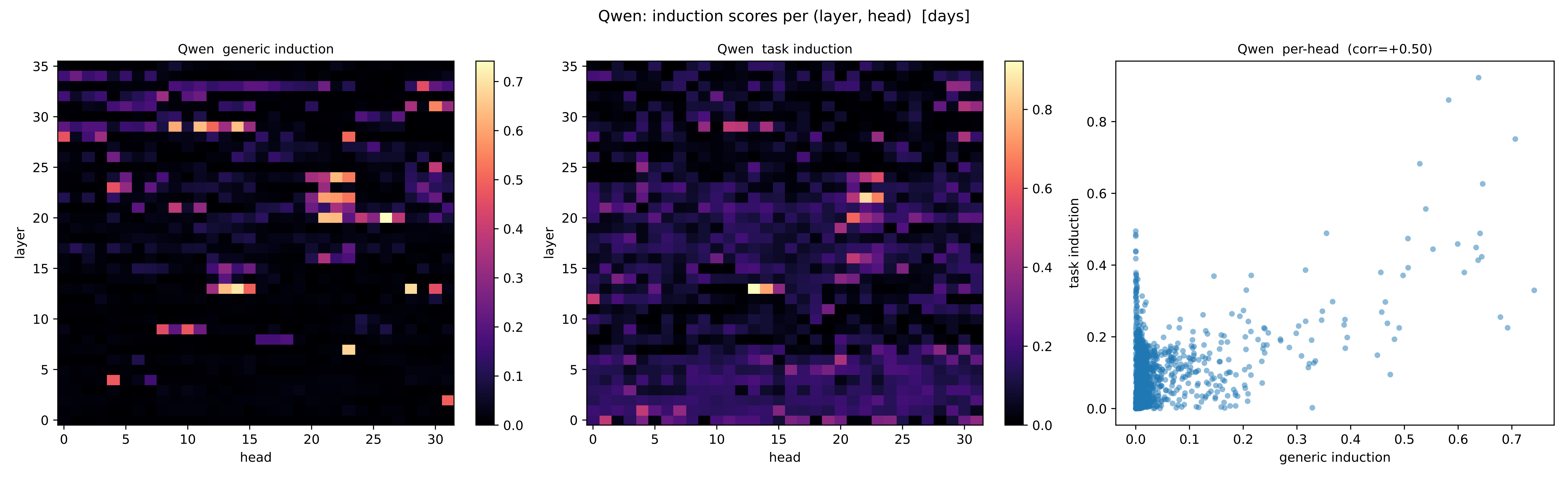


Llama: attention PATTERN of the top induction heads
Llama GENERIC: head L15H30 on repeated-random seq
bright cells on the red line = induction stripe
induction target (key = query−L+1)
query position
key position
Llama TASK: head L16H20 on a walk
red rings = current-node → prior-occurrence successors
prior-successor targets
query position
key position

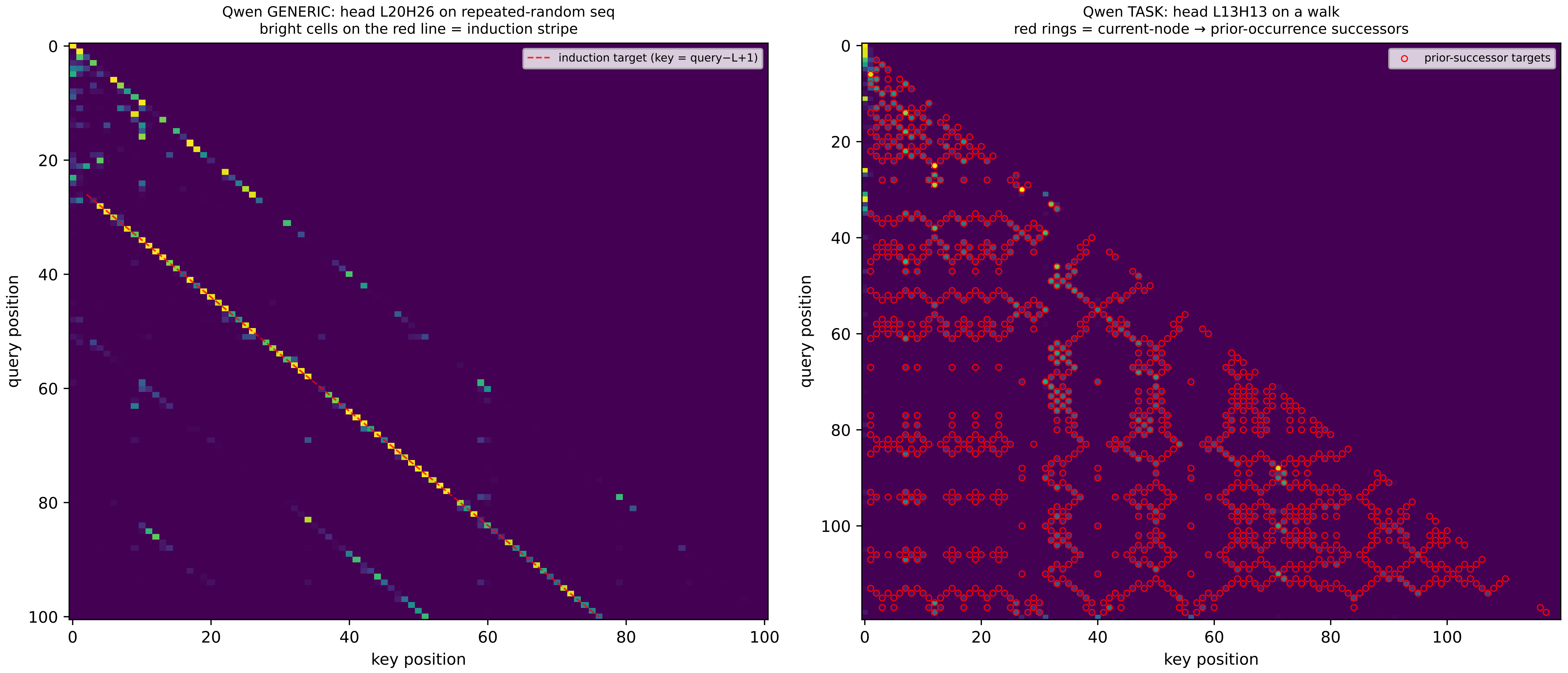


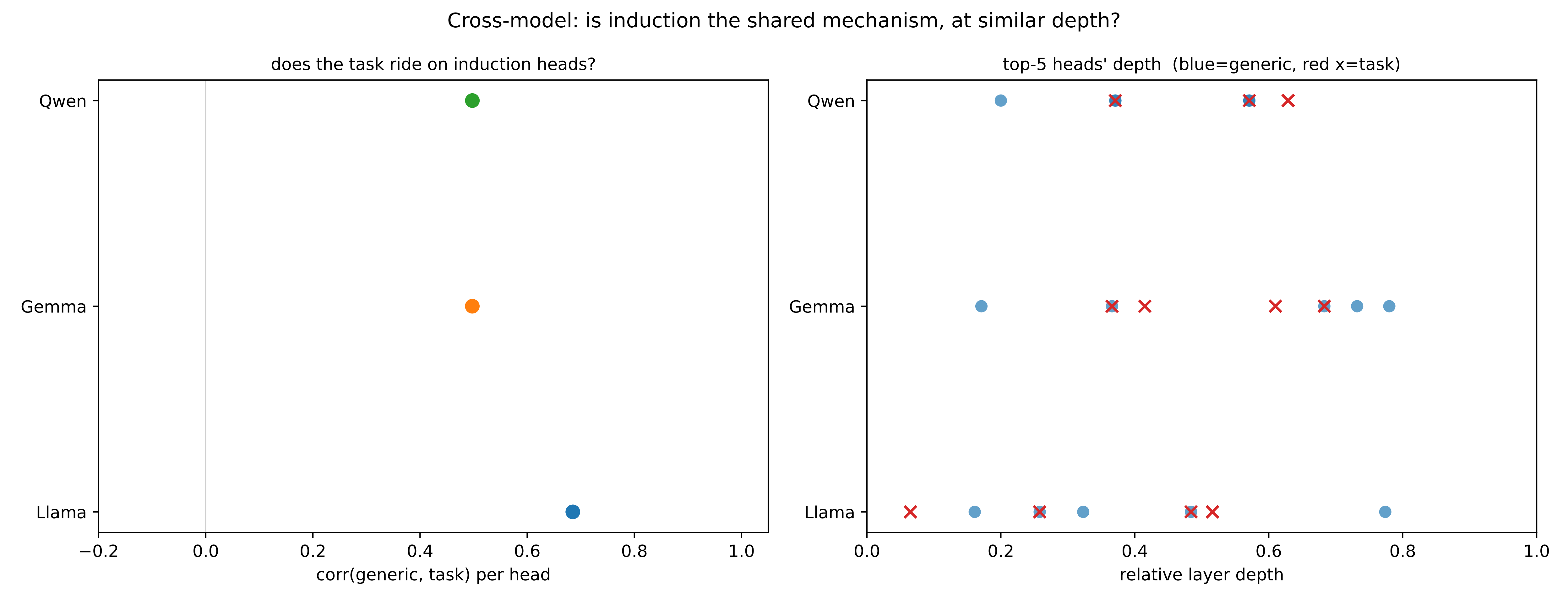
Gemma: induction scores per (layer, head) [days]
Gemma generic induction
Gemma task induction
Gemma per-head (corr=+0.50)
layer
head
task induction
generic induction


Gemma: attention PATTERN of the top induction heads
Gemma GENERIC: head L15H3 on repeated-random seq
bright cells on the red line = induction stripe
induction target (key = query−L+1)
query position
key position
Gemma TASK: head L28H2 on a walk
red rings = current-node → prior-occurrence successors
prior-successor targets
query position
key position


Qwen: induction scores per (layer, head) [days]
Qwen generic induction
Qwen task induction
Qwen per-head (corr=+0.50)
layer
head
task induction
generic induction

Qwen: attention PATTERN of the top induction heads




Cross-model: is induction the shared mechanism, at similar depth?
does the task ride on induction heads?
Qwen
Gemma
Llama
−0.2
0.0
0.2
0.4
0.6
0.8
1.0
corr(generic, task) per head
top-5 heads' depth (blue=generic, red x=task)
relative layer depth